

TAPAS BARMAN

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EDUCATION

Indian Institute of Information Technology, Lucknow

MSc Data Science — SGPI: 8.05

Lucknow, UP

2024 – 2026

EXPERIENCE

MLOps Intern

Dec 2025 – Mar 2026

Aaizel International Technologies Pvt. Ltd.

On-Site

- Built production-grade **ETL & ML pipeline** (**Airflow, S3**) automating ingestion and processing of **3GB of GEOS-FP geospatial data every 3 hours**; implemented automated retraining, drift monitoring, and CI/CD via **MLflow, DVC, Evidently**, and Jenkins.
- Developed **ML backend** for **NER, LLM embeddings, vector search**, and clustering, ingesting **900+ RSS feeds/scrapers** (150 Indian, 750 international) into a **Supabase-backed knowledge graph** for intelligence extraction and entity resolution.

Computer Vision & AI Research Intern

Jul 2025 – Dec 2025

Indian Institute of Technology Bombay (IIT Bombay)

Remote

- Worked on a platform for **digital preservation and archiving** of Siddha literature across **25 Indian languages**, through OCR-based text digitization and proofreading.
- Replaced Tesseract with **GPU-based EasyOCR**, achieving a **3x speedup** and **18% error reduction**.
- Built table detection and reconstruction using **YOLO + Transformers**, improving accuracy by **22%**.

PROJECTS

Route Pilot | Python, Playwright, LangGraph, DeepEval, Telegram Bot

GitHub

- Built hybrid Planner–Actor–Evaluator browser agent (**Playwright**) for autonomous bus search/booking; validated via **DeepEval** across **78 golden tests (98%+ pass rate)** and cut token cost **98.9%** (45K → 405 tokens/query).

RAG Evaluation Suite | Python, LangChain, ChromaDB, Gemini, DeepEval

GitHub

- Built two-stage **RAG pipeline** (**ChromaDB + MS-MARCO reranker**) with a **DeepEval** regression suite (Faithfulness, Answer Relevancy, Contextual Precision/Recall), achieving **100% guardrail pass rate** at **\$0.00018/query**.

LLM Fine-Tuning: Support Agent | Python, PyTorch, Hugging Face, Unsloth, LoRA/QLoRA

Colab

- Fine-tuned **Mistral-7B-Instruct** on **27K samples** via 4-bit QLoRA with Unsloth, reducing VRAM by **70%** and speeding training **5x** using sequence packing, response-only loss masking, and NEFTune.

Visa Prediction – End-to-End MLOps Pipeline | Python, Scikit-learn, MongoDB Atlas, Docker, AWS, FastAPI

GitHub

- Production-grade **MLOps** pipeline for visa approval forecasting with automated training, model selection, and **CI/CD** deployment on **AWS EC2/ECR** via GitHub Actions.

TECHNICAL SKILLS

Languages: Python, SQL

Data & APIs: Pandas, NumPy, FastAPI, REST-APIs, SQLite

ML & Deep Learning: PyTorch, Scikit-learn, Hugging Face Transformers, LoRA, QLoRA

Generative AI: LangChain, LangGraph, RAG, Reranking, DeepEval, LLM Evaluation, Guardrails

Infrastructure & DevOps: Linux, Git, Docker, AWS (EC2, S3, IAM, RDS, Lambda)

Data Engineering & MLOps: Apache Airflow, Apache Kafka, MLflow (Tracking & Registry), ETL & ELT, Evidently AI

ACHIEVEMENTS

- Meta Hacker Cup 2025: Secured **global rank under 2000**.
- TCS CodeVita 2025: Achieved **rank 585** out of 20,540 participants.
- OpenAI Open Model Hackathon: Successfully participated and submitted a project.
- IIT-JAM 2024: Secured **All India Rank (AIR) 1200** in Physics.
- LeetCode: Solved **350+** submissions; active competitive programmer.

LEADERSHIP EXPERIENCE

Community Lead – Web and Coding (WnC) Wing

Aug 2025 – Present

Aether – The Data Science & AI/ML Club of IIIT Lucknow

- Organizing coding sessions, competitive programming contests, and technical workshops for MSc, MTech, and PhD students.